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ALIGNMENT

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Introduction

Purpose

The purpose of this document is to help Information Technology groups adopt their new mandate and roles, given the advent of the new systems. The document captures the new vision for the groups, their new organizational structure, their new human performance support systems, and their new learning requirements. The Information Technology groups need to make sure of their integration with the overall business strategy and reflect their new capability.

Purpose of New Information Technology Alignment Strategy

More specifically, while facilitating the migration from the legacy to the new systems, the new information technology alignment also contributes to the following needs:

- Make sure the Information Technology groups sustain the technical gains initiated with the implementation
- Support retention of staff
- Identify Information Technology career development paths
- Determine ways to position Information Technology as a proactive area worth investing in, not only as a cost centers
- Build enthusiasm for the opportunities the changes will bring
- Involve key employees in planning the transition
- Re-skill the groups for the new technologies

Overview

This document includes the following components:

Component	Description
Information Technology Groups' Mission	Describes the updated charter for the Information Technology groups
Current Information Technology Human Performance Management Systems	Guides the analysis of current performance support systems in the Information Technology groups
New Information Technology Human Performance Support Systems Requirements	Identifies the human performance support systems requirements for the new Information Technology roles and the approach to meet those requirements
Information Technology Human Performance Support Systems Recommendations	Lists the recommendations for the development of the Information Technology groups' new human performance support systems, the required deliverables, the revisions to the Human Resource Information Systems and the roll out

Component	Description
Information Technology Human Performance Support Systems Materials	Includes new job descriptions, new organizational chart
New Information Technology Work flows, job structure and organizational structure	
New Information Technology Job Performance	
New Information Technology Job evaluation and classification	
New Information Technology Performance support practices, procedures, tools, systems	Includes new competency models, productivity expectations and other performance metrics, human resource allocations
New Information Technology Job-person matching	
Information Technology Human Performance Support Systems Measurement	Includes grade levels, classifications, compensation ranges
Information Technology Learning Plan	Includes deliverables to guide managers when they interview, select, appraise performance, set performance goals, provide feedback, plan career development, reward and recognize
Appendix A	Includes new skills profiles, learning plan, succession plan
Appendix B	Includes instruments as determined by the measurement methods recommended in the HPSS strategy
Information Technology Learning Plan	Description of learning events for Information Technology job incumbents
Appendix A	Sample of Common Information Technology Job Roles
Appendix B	Traits for Information Technology Competencies

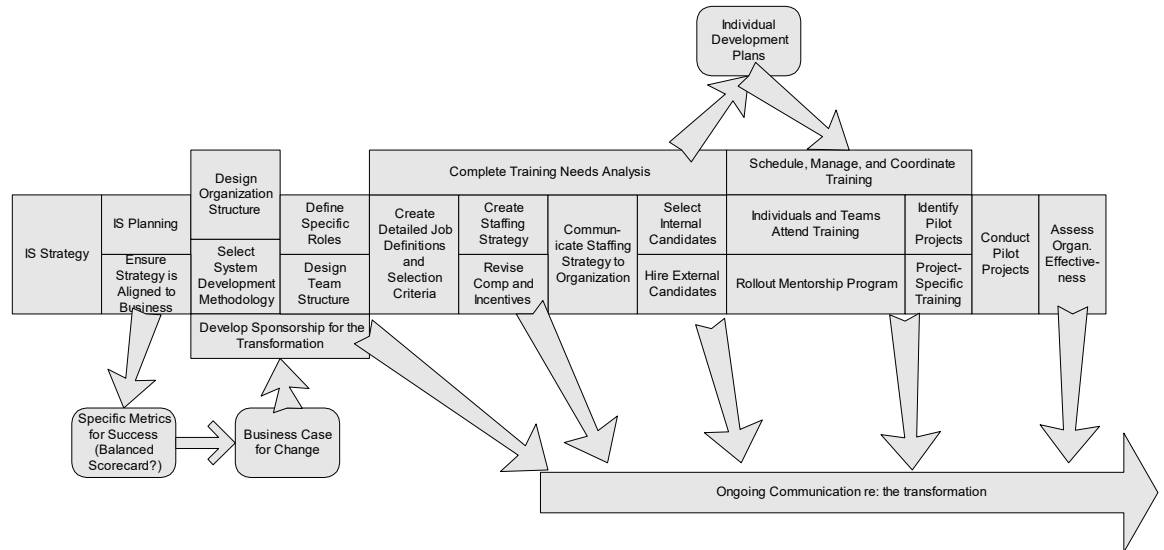
Key Considerations

To prepare for the conversion of Information Technology groups either from their current computing/technology environment to one that supports the new technology approach, the following considerations need to be addressed. This listing is not exhaustive. It is only intended as guide. The sections in this document detailed the key components of the alignment strategy.

- What's the current mode of operation? Who is on the staff? How many are on the staff? What are their skills? How does the current mode of operation support the current Information Technology vision? Do you provide user support? How is legacy system maintenance addressed? How is current learning accomplished? How are system changes introduced?
- What's the future mode of operation? What is the staffing plan? What are the re-skilling expectations? How does the future mode of operation support the future Information Technology vision? What business objectives does the future mode of operation support? Who establishes the budget? Who administers the budget? Does the business expect productivity gains? How are they measured? Once achieved, how will they be maintained? How will changes in business objectives be monitored? What is the staff retention strategy?
- How does the current mode of operation compare to the future mode of operation? What's the magnitude of the approach? What is the technology/platform approach? What are the effects on current staff – re-place, re-skill, reduce, outsource? What are the learning implications? What should be the ongoing user support strategy – help desk, learning databases, security, legacy system support? How will new technology be supported – Technical Assistance Requests (TAR)? Pending completion of the technology approach, is there a need to create interim work-around solutions - learning events, alternative data sources, revised/new business processes, etc.

- What's the transition delta between current mode of operation and future mode of operation? What are the changes in Information Technology staffing requirements? What are the skills requirements? Who manages the learning events budget?

Alignment Approach



Information Technology Groups' Mission

Technology Transition

When an organization takes on large technology changes, the structure must be in place to both support the new technical environment and help the user groups make the transition as smoothly as possible. This requires a well designed strategy and well executed plan that both retains the critical employees and prepares them for the future role they will play.

Through <Project Name>, <Company Long Name> is transitioning from:

From...	To...
Mainframe	Internet
Client/Server	Internet Computing
Mainframe Operating System	UNIX
Centralized Data Access/Report Generation	Open Data-Warehouse
Interwoven Complex of interfaced systems	Single integrated system

All of these changes affect the Information Technology groups profoundly and deeply in the way they work, in what they have to know, and how they support the corporate strategy. The retention and effectiveness of this team is a major success factor for the implementation.

Alignment of Information Technology Mission with Business Strategy

This section focuses on shared understanding of how the goals for the Information Technology groups create business benefits, how they are linked to the corporate strategy, and how success will be measured.

Information Technology Groups' Mission	Corporate Mission	Corporate Vision

Priority #	Business Objectives for Information Technology groups	Benefits	Measurement

Benefits of Information Technology Alignment

By aligning the Information Technology mission to the business strategy, the organization will benefit from the following:

- Information Technology groups skilled for the new environment and able to sustain the technical gains
- Information Technology groups better able to help users through their change

- Perception of the Information Technology groups as a true value added player in the organization

Current Information Technology Human Performance Support Systems

List of Roles Impacted

The following are the roles impacted by the new Information Technology Management process and procedures; indicated for each is the customer base (business unit/department served by the role if known) and the related procedures that defines how to play the role:

Information Technology Management Process Procedure	Role	Organization Base (Business Unit/Department)	Comments

Change Context

To make sure the human performance support systems strategy is consistent with the business drivers and overall Information Technology groups' strategy, it is important to understand the human resource and performance management context. This clarifies the groups' thinking into the human resource and performance management direction and implications associated with the new mission. These insights provide the basis for deriving the performance support objectives for the new Information Technology mission.

Business Motivation	Questions
When upgrading operations	What are the new expectations for the Information Technology groups' performance? What experience does the department have in upgrading operations, for example, moving from mainframe to client/server?
When expanding into new areas of business or service	Are employees given an opportunity to learn the new business, or are individuals hired with the appropriate skills and experience in the new business? How is it determined whether performers and managers have the skills and abilities they need?
With new skills that are required	Are existing employees being laid-off and new ones hired? Are current employees selected for reskilling and redeployment?

Current Information Technology Human Performance Context

Listed below are the elements of the Information Technology human performance support systems currently available. Their inventory is the first step to determine the gap between current and future requirements.

Performance Area	Currently Available	Reference
Work flows, job structure, organizational structure current job descriptions and work flows, and an estimate of their validity job and departmental organization recent reorganizations that might relate to the current change effort project team structure and philosophy system development and maintenance approaches and methods		Policies and procedures manuals Orientation handbooks Learning manuals Organizational chart Existing job descriptions Equipment specifications Union contracts Business Process Reengineering documentation System development methodology handbooks Project definition documents
Job Performance competency models, their availability and use existing methods/processes for determining performance behaviors existing methods for determining productivity expectations and resource allocation productivity measurement tools		Past performance review Current productivity metrics Management handbook Budgeting procedure
Job evaluation and Classification current grade levels current job classifications and corresponding content compensation ranges		Existing job descriptions and postings Human Resource policies and procedures (compensation, etc.,) Union contracts
Performance Support Practices, Tools and Systems performance management, including goal setting and performance planning, feedback and performance appraisal rewards and recognition programs recruitment methods/processes/tools selection methods/processes/tools outplacement programs career planning methods/processes/tools quality improvement programs		Human Resource policies and procedures manuals Orientation handbooks Union contracts
Job-Person Matching current skills requirements and competency profiles current learning plans and offerings succession planning programs		Learning manuals Learning records Past performance reviews

New Information Technology Human Performance Support Systems Requirements

Assessing the Relevancy of Current Information Technology Human Performance Support Systems

Work Flows, Job Structure and Organizational Structure

The following questions guided the identification of requirements related to work flows, job structure and organizational structure:

- What are the tasks required by the new Information Technology mission?
- Do the existing jobs cover all the tasks required by the new Information Technology mission?
- Are job tasks grouped in the most efficient ways? In ways that support the objectives of the department
- Do job structures require that performers
 - ◊ use multiple skills (skill variety)
 - ◊ complete a whole piece of work, that is, produce a visible outcome (task identity)
 - ◊ perform tasks that have an impact (task significance)
 - ◊ receive information at the point of performance (feedback)
 - ◊ determine how to carry out their jobs (autonomy)
 - ◊ cooperate with others (teamwork)
- Are the Information Technology work flows organized in the most efficient ways? In ways that support the objectives of the department?
- Are there unnecessary overlaps in responsibility for tasks?
- Are variances (changes in the state of job outputs) controlled close to the point of origin?
- Is control of variances defined in the job structures?
- Is there rotation of responsibilities and work content among performers?
- Do departmental boundaries reflect the natural breaks in the new or changed Information Technology work processes?
- Do technology (complexity), territory (distance), and time (24 hour processes, etc.) impose limitations on work as it is currently organized?
- Is there interdependence among departments for needed resources?
- Are there too many participants in work flows to promote interaction/cooperation (> 14-15 performers)?
- Do performers have equal access to equipment, etc.?

Job Performance

The following questions guided the identification of requirements related to job performance:

- Do existing competency models fit the job descriptions required by the new or changed Information Technology work processes?
- Were competencies identified in the assessment that are not reflected in the competency models drafted in the New Performance Specifications for Each Job Group?
- Do existing productivity metrics describe the outputs of the new Information Technology work processes?
- Do existing productivity metrics allow for diagnosis and mid-stream corrections? Are there lagging (for example net profit) and leading (for example lead times, quality) indicators?
- Do existing productivity metrics measure development of competence and potential?
- Will existing productivity metrics provide information that is relevant to the objectives of the new technology implementation?
- Are the existing metrics counterproductive to the objectives of the new technology implementation?
- Are there hidden culture-driven metrics that may impair world-class performance?
- Does the existing process for setting productivity metrics involve both strategic alignment and bottom-up input?
- Are existing productivity metrics
 - ◊ used to control operations and improve process
 - ◊ produced regularly, not simply on request
 - ◊ believed and understood
 - ◊ described at the most significant level of detail/in the most significant cuts
 - ◊ complete enough to provide a whole picture
 - ◊ accompanied by plans for periodic updating
 - ◊ reported?
- Is productivity measured at the right level of aggregation (individual, team, department, organization)? At multiple levels of aggregation?
- Do productivity metrics include customer data?
- Are there multiple measures of productivity (for example quality, delivery, time, cost)?
- Are human resource allocation data and models appropriate for the new or changed work processes?

Job Evaluation and Classification

The following questions guided the identification of requirements related to job evaluation and classification:

- Will existing job classifications or grade levels match the new or changed job descriptions?
- Does compensation need to be reevaluated to reflect any changes in job descriptions?
- Is existing job compensation internally and externally equitable?
- Will compensation need to be adjusted for different standards of living? (Are compensation levels to be centralized or decentralized?)
- Does the number of classifications/grade levels (level of bureaucratization) reflect the objectives for the department?
- Is the job content comparable for jobs within the same job classifications or grade levels?
- Does the existing method of job evaluation (factor comparison, points rating, person-to-person comparison, etc.) determine the relative worth of a job in ways that reflect the objectives of the department?
- Are the relative weights for the factors or points used in job evaluation aligned with the values and objectives of the department (for example, educational level more important than experience)?
- What are the behavioral consequences for the existing method of job evaluation (for example, increased competition in the case of person-to-person comparison)?
- Is the existing method of job evaluation perceived as fair?
- Are all legal implications for compensation and job evaluation understood and addressed?

Performance Support, Practices, Tools and Systems

The following questions guided the identification of requirements related to performance support, practices, tools and systems:

- Do the existing performance support tools (for selection, performance management, etc.) reflect the new job and organizational structures, competency models, productivity metrics, etc.? Do they need to be maintained, revised, discarded?
- Do new performance support procedures, tools, and systems need to be added?
- Do the performance support practices, tools, and system link performance to the new or changed strategic goals and objectives?
- Is managers' current reskilling in performance support appropriate for the new performance support procedures, tools and systems?

Performance Review:

- Does the content of performance reviews fit the behavioral objectives of the new or changed work processes?
- Is there a mechanism for frequent feedback on performance?
- Is there a participative process for establishing performance objectives and goals?
- Does the performance review process focus on both accomplishments and areas to be developed?
- Do employees have a means for questioning performance reviews?

Promotion/Transfer/Career Development:

- What career development options are available? Are they communicated?
- Does the content of the career development options align with the new or changed development needs, given the new competencies required and career paths available?
- Do the career development options encourage experience and exploration outside the job incumbent's current work content?
- Is career development rewarded? Funded?
- Do career development options make use of outside resources available to the organization?
- Are there programs for external as well as internal Human Resource planning (for example, sponsorship of relevant community university programs to promote the development of the job candidate pool)?
- How is career development linked to the Human Resource Information System (HRIS)?
- How are career development options evaluated?

Rewards and Recognition:

- Do the existing rewards and recognition practices support behaviors that align with the objectives of the change effort? Do they need to reward maintenance or improvement? Potential or actual performance? Behaviors or results?
- Are rewards and recognition incentives based on the correct level of aggregation (individual, team, department, organization), given the new or changed work processes?
- Do/will all target job groups have the materials, equipment, working conditions, etc. needed to make sure that performance-based incentives are fair?
- Are rewards and recognition incentives made at the appropriate intervals? Are there short-term and long-term awards that align with strategic goals?
- Is the incentive program cost-effective?
- Is the incentive program perceived as competitive and desirable to job incumbents?

Recruitment, Selection, Termination

- Will recruitment and selection procedures reach and attract the right job candidates, given the new or changed Information Technology work processes?
- Is recruitment both an internal and an external process?
- Is there a formal termination process? Does it reflect legal considerations?
- Is the termination process linked to the performance review process?
- Does the basis (bases) for termination reflect the objectives of the department?

Job Person Matching

The following questions guided the identification of requirements related to job person matching:

- What options exist for succession planning and at what levels? How are they linked to career development? To the Human Resource Information Systems?
- Is succession planning competency-based?
- Are all the necessary skills for the new or changed work processes covered in the target job groups? By the relevant job groups?
- Do existing learning options cover all the skills gaps?
- Do incumbent skills sets map efficiently to their existing or changed job responsibilities?
- Do existing learning options meet the logistical requirements and preferred learning styles of the target job groups?

Listing of Performance Support Needs

Based on the findings from the current Information Technology human resource and performance management context and the assessment of its relevancy to the new performance environment, we have identified the following list of performance support needs.

Findings	Performance Support Needs
Work Flows, Job Structure and Organizational Structure	
	Example: The following jobs/roles need to be realigned, the following jobs/roles remain as they are, the following jobs/roles need to be created or designed, the following jobs/roles will go away, the following departments need to be merged or reconfigured, etc.

Findings	Performance Support Needs
job tasks/responsibilities job interdependencies work flow (inputs/outputs) reporting relationships customers and suppliers machines, tools, equipment, work aids (MTEWA) used scope of authority autonomy and process control (TQM) decision making process organizational layers/hierarchy business processes department boundaries redundancies inefficiencies working conditions teams work procedures	
Job Performance	
	For example, for this role (or group of role), the following competencies need to be added, reinforced, or removed for each role, the following performance expectations need to be augmented, reinforced, the following resource allocation will be...
consequences of work behaviors performance criteria productivity metrics algorithms, consistent behavior patterns of superior performers behaviors that add value, given the objectives of the technology-induced change effort customer and supplier needs/satisfaction (to guide customer-focused behaviors) human resource allocation	
Job Evaluation and Classification	
	For example, the following classification system needs to be changed to realign compensation
task attributes (importance, variety, difficulty) current job classifications grade levels rating factors union guidelines information relevant to criteria for standard job classifications (for example, government codes) scope of responsibility competencies required degree of control supervisor's style compensation ranges compensation benchmark in industry and market raises	
Performance Support Practices, Tools and Systems	
	For example, the following tools, procedures and systems are still adequate, the following need to be changed or removed, the following need to be designed or added, etc.

Findings	Performance Support Needs
goal setting/performance planning hiring promotion performance feedback recruitment selection learning termination staffing career development rewards and recognition performance review policies (ethics, issue resolution, grievance, etc.)	
Job-Person Matching	
	For example, the following learning events are recommended, for these future role incumbents, the following learning programs need to be developed, the following new skills need to be acquired, the following succession plans need to be redesigned, etc.
knowledge, education level skills aptitudes orientation (attitude, preferences, etc.) learning needs succession planning	

Prioritized Job/Role and Performance Support Needs

Jobs/roles whose performance support needs will form the highest priorities in the approaches are those who have a high impact on the success of the change effort and whose current performance support infrastructure requires the most development to reach its optimal state. The next priorities are those jobs/roles whose impact is high, but whose current performance support infrastructure requires less development. Jobs/roles whose impact is medium become priorities after all job groups whose impact is high have been considered, regardless of the relative weight of their performance support gaps.

We prioritized performance support needs for each Information Technology priority job/role by considering both the value they add to achieving the objectives of the Information Technology groups, as well as the extent of their gap between existing and optimal states. The highest priorities are those that add the most to the achievement of the objectives and who also need the most work. The value they bring to the change effort takes precedence over the amount of work needing to be done to optimize them.

The following jobs/roles have the highest priorities and the support needs have the highest impact on the success of the Information Technology mission:

Priority Roles	Priority Support Needs

Objectives of New Information Technology Human Performance Support Systems

The objectives of the new Information Technology performance support processes are:

- Enhance employee commitment to the new departmental objectives and human resource processes.
- Explain new Information Technology strategies. Employees must understand the discrepancy between their current behaviors and the consequences of these behaviors and new expectations.
- Personalize the strategies.
- Implement Human Resource programs that are consistent with the new Information Technology strategies and performance support systems.
- Enhance personal accountability.

These objectives are consistent with the following needs of our users/performers:

- A clear sense of direction.
- Opportunity to participate in goals and standards for performance.
- Timely, honest, meaningful feedback (both what is effective and what can be improved) as they do their work.
- Immediate, meaningful, sincere reinforcement for their efforts.
- Coaching and assistance on ways to learn and improve.
- Treatment that involves fairness, sincerity, respect.
- Opportunity to understand and possibly influence decisions that affect them.

Specific Objectives per New Performance Process

The following captures objectives specific to the new performance process:

Performance Process	Objectives
Work Flows, Job Structure, Organizational Structure	
Work flows/team	Designing the flow and organization of work may involve some or all of the following objectives: - Align the work flow to optimize the new or changed Information Technology work processes. - Improve response time and quality by enabling control of the variances in the process where they occur. - Organize teams where performance and output would benefit. - Facilitate career development by providing job incumbents with opportunities to develop more skills, through job rotation and different assignments. - Increase employee identification with the end product or service by designing jobs that involve complete processes and meaningful tasks. - Enlarge (provide more authority) and enrich (provide more variety of tasks) jobs - Reduce the number of process steps, and thus length of the process, to speed time to completion/time to market. - Increase the competencies of employees by creating less rigid job content and broadening opportunities to share responsibilities and tasks.

Performance Process	Objectives
Job descriptions	Accurate, up-to-date job descriptions that describe information relevant to the objectives of the Information Technology groups will form the foundation for many of the other performance support areas. Their objectives are to: Help managers determine whether all responsibilities are covered and allocated correctly. Clarify the ownership of responsibilities. Help incumbent understand what is expected of them. Focus career development and performance appraisals. Define criteria for new job grades, wage and salary structure. Identify skills requirements. Guide productivity metrics. Determine needs for hiring, learning, and outplacement guidelines. Guide the organizational structure. Every target job will have an up-to-date job description.
Organizational structure/ reorganization	Organizational structure, like work flows, seeks to optimize the organization of the new or changed Information Technology work processes. In addition to the objectives described under work flows, reorganizations or adjustments in organizational structure may seek to: Reflect changes in job structure and competencies, time requirements, need for coordination, centralization/ decentralization of operations, and feedback time, as mandated by the new work processes. Provide more autonomy and reduce dependence on other departments/individuals for resources, etc. Provide individuals who must coordinate and cooperate with the physical proximity needed to do their job. Increase identification and quality control by localizing entire processes such that one group is responsible from start to finish. Eliminate inefficient redundancies Reduce the number of jobs, thus length of process, to speed time to completion/ time to market
Work Flows, Job Structure, Organizational Structure	
Competency Models	Competency models seek to: Detail the knowledge, skills, aptitudes, and orientation required for job incumbents to be successful given the new or changed work processes. Indicate the behaviors and thought algorithms common to superior performers. Allow the interview and selection process to identify those individuals most likely to excel in the work, by focusing on past performance and behaviors in addition to experience and expertise.
Productivity metrics/productivity measurement system	Some of the possible objectives for productivity metrics are to: Encourage desired behaviors (what gets measured gets done) and/or demonstrate what is valued (what gets measured is perceived as important). Suggest trends and possible improvements while "in process," that is, Operational control. Estimate future capability through the measurement of strategic investments (in people as well as in capital). Guide expectations for acceptable or exceptional performance, for possible use in performance reviews and the determination of rewards and recognition. Demonstrate and report achievement. Suggest areas that require attention (areas where productivity lags). Determine resource allocation, that is, how many FTEs (Full Time Equivalent) for each job.
Job Evaluation and Classification	
Job measurement/base pay	Some of the possible objectives for job measurement are to: Price jobs according to their relationship to strategic objectives for the Information Technology groups, by developing new criteria that are in line with the objectives. Realign compensation to reflect the new responsibilities and competencies required of performers. Compete successfully in the job market for those candidates who fit the job requirements as dictated by the new or changed work processes and the objectives of the Information Technology groups. Develop a job measurement process that encourages new values (for example, points rating, which does not encourage individual competitiveness in the way that ranking does)

Performance Process	Objectives
Classification/grade levels	Possible objectives for job classification/grade levels are similar to those of job measurement: Align classifications and grade levels with what is valued in the Information Technology groups Encourage the acquisition and competencies that are in demand, through the systematization of rating or ranking factors that reflect the values of the Information Technology groups Dilute the bureaucratic or hierarchical configuration of the organization to promote fluidity and flexibility
Performance Support Practices, Tools and Systems	
Recruitment Selection Orientation Transfer	Recruitment, selection and orientation strive to: Reflect the job descriptions and competency models, in order to guide the staffing process in accordance with the values and objectives of the Information Technology groups Identify the appropriate sources for potential employees In the case of orientation, inform employees of the department's mission, vision, and strategic goals, as well as details of the performance support system.
Outplacement Termination	Although the aim of performance support is the development of job incumbents, performance support tools to ease termination and facilitate outplacement may also be made available. Attention is paid to the emotional needs of the performer, as well as assistance in the transition
Goal setting and action planning	Goal setting objectives may include: Communicate to employees the criteria by which their performance will be measured. Allow for the joint development of plans for achieving performance goals. Build employee commitment to the achievement of goals that are in line with the strategic objectives of the department and whole organization. Demonstrate to the employee the value of their job in contributing to the achievement of overall organizational goals. Recognize the employee's career goals and identify resources and opportunities available for furthering them
Feedback	Feedback processes are key to performance management. They strive to: Provide the employee with information on how their performance is meeting, or missing, the target goals Allow the employees to adjust or strengthen performance in time to meet performance goals Provide feedback that is: ongoing, timely, frequent, reinforcing, objective, positive
Performance review	With ongoing feedback in place, performance reviews become opportunities to: Celebrate goals met Review lessons learned Plan the next performance cycle Discussion of bonuses, raises, and other financial aspects of the job are left to another occasion.
Career planning/employee development/transfer	Career planning strives to: Identify ways to help individuals understand corporate changes in career opportunities and job expectations. Identify key experiences important to firm and important to individual's career development. Evaluate managerial skills and functional knowledge in relation to emerging opportunities. Enhance self-confidence, willingness to take reasoned risks, and need for achievement. Enhance problem-solving and relationship skills.
Raises and promotion	Objectives for raises and promotions can include: Base raises and promotions on performance, rather than the simple passage of time. Reward <i>how</i> results are achieved, as well as <i>if</i> results are achieved. Promote the values of the department given the objectives of the Information Technology groups (for example, if the raise pool is fixed, such that one person's above average raise means less for someone else, competitiveness may be enhanced at the expense of teamwork).

Performance Process	Objectives
Team celebrations	Celebrations that honor group achievements of desired objectives strive to: Celebrate group accomplishments in ways that are personally meaningful to the employees. Strengthen cooperation as well as commitment. Increase identification with the goals of Information Technology groups.
Individual contributor awards	Individual contributor awards may have the following objectives: Highlight certain individual behaviors for possible emulation Enhance the sense of personal responsibility and contribution Reward individual contributions to team goals
Non-monetary awards	Non-monetary awards may seek to: Reward behavior and accomplishments in ways that are relevant to the employees. Provide rewards and recognition while maintaining cost-effectiveness of the incentives program. Enhance the market competitiveness of the position. Increase the perceived desirability of the organization to prospective employees. Reinforce the corporate culture.
Job-Person Matching	
Succession planning	Succession planning processes are optimally available for all levels of the organization. Succession planning is designed to: Develop individuals who possess the requisite skills to lead in the environment created by the Information Technology groups. Minimize disruption to the department's strategic mission and vision during the change effort by providing individuals who are already "on board" and prepared to step into positions once they become vacant. Plan leadership resource strategies for the short and long term. Retain individuals with valuable experience while ensuring that they have the required cross-functional skills to succeed at higher levels. Realize the potential of especially promising new graduates
Learning	Reskill employees to meet the new demands corresponding to the Information Technology groups' objectives.
Job assignment	If there has been a skills assessment, performance support may include tools that map employees' skills and interests to the requirements of the jobs to: Identify which employees are most suited for which new jobs Align current and future resources in the most mutually beneficial ways, for example, ways that increase productivity and opportunities to succeed Identify jobs for which current employees need additional learning Determine best resourcing options for vacant jobs

Information Technology Human Performance Support Systems Recommendations

Work Flows, Job Structure, and Organizational Structure

The following documents recommendations related to work flows, job structure, and organizational structure:

Job Performance

The following documents recommendations related to job performance:

Job Evaluation and Classification

The following documents recommendations related to job evaluation and classification:

Performance Support Practices, Tools and Systems

The following documents recommendations related to performance support practices, tools and systems:

Job-Person Matching

The following documents recommendations related to job-person matching:

Effectiveness Measurement

To make sure of the effectiveness of the Information Technology human performance support processes deployed, we recommend the following measurement approach:

Measurement Approach	Recommended Approach
Performance support deployment scorecard criteria	
Performance support systems scorecard criteria	
List of measurement methods	
List of measurement deliverables	
List of measurement events	

Required Materials

In light of the above recommendations, the following materials will be required for the prioritized roles:

Prioritized Roles	Required HPSS Materials

Human Resource Information Systems Implications

Each of the performance support process has implications on the human resource information system. When performance support processes have been defined, we will determine which processes link to the Human Resource Information Systems and to what end. We will then review the configuration to determine any recommendations for revisions to the system. At this point, we have identified the following:

- ...
-
-

Roll Out Requirements

The roll out requirements involve communication and learning on how to use the tools developed for performance support, including initial goal setting and planning sessions, description of new job opportunities or duties, description of the performance management process, and clarification of criteria for performance measurement.

The following communication and learning events aimed at <relevant Human Resource personnel, managers/supervisors for the target job/role groups, users and other stakeholders> need to take place:

Target	Content	Communication Events	Learning Events
Human Resource Personnel			
Managers/Supervisors			
Employees			
Other:			

Information Technology Human Performance Support Systems Materials

New Information Technology Work Flows, Job Structure and Organizational Structure

Job Descriptions

Job descriptions include the following information:

- Job title
- Purpose (objectives the job is designed to meet)
- Job summary (what distinguishes it from other jobs)
- Authority and accountability (reporting relationships)
- Primary and secondary responsibilities (ranked)
- Description of tasks
- Date of last revision

They also include such information as:

- Grade level
- Traits or attributes required of the job incumbent (educational level – minimum and preferred, experience, learning, skills, etc.)
- Union affiliation

New Information Technology Job Performance

Competency Models

Competency models show a hierarchy of competency information:

- Cluster
 - ◊ Competency 1
 - ◊ Competency 2

Competency are clustered by:

- Dictionary cluster (for example, achievement-related, power-related, cognitive, interpersonal (affiliation), self-management)
- Job task or responsibility (for example, budgeting)
- Logical sequence in time

Competency models also show:

- Job responsibility by competency matrix showing which competencies are needed to de each part of the job
- Algorithms

Productivity Metrics

Sample of metrics applicable to the Information Technology groups include:

- total cost of ownership: Includes the total cost to design, develop and maintain the system. Estimates on yearly maintenance are used to calculate.
- help desk response time: how quickly do support calls get resolved
- customer satisfaction: how satisfied are customers with the system
- software defect rates: how many bugs are in the implemented software
- problems cleared by the help desk: number of issues closed

Wilson describes the qualities of effective measures with the acronym PROACT:

Positive	Phrased positively (indicating progress) rather than negative (pointing out obstacles, failure)
Reliable	Consistently accurate and verifiable
Objective	Data are collected from multiple sources
Active	You must have to do something to achieve it; it cannot happen without effort
Controllable	Within influence of performer
Timebound	Should indicate acceptable timeframe for completion

New Information Technology Job Evaluation and Classification

Depending on the method chosen in the approved recommendations, the job evaluation process involves:

1. Determining the factors that are value-added give the objectives for the new technology implementation.
2. Devising ratings scales or levels that indicate the presence (and where appropriate degree of presence) of these factors.
3. Clustering positions into grade levels or rankings according to overall scores for the ratings and levels
4. Pricing the grade levels

New Information Technology Performance Support Practices, Procedures, Tools and Systems

Performance Review

Performance management deliverables consist of manager's guides, and performance review instruments. The performance review template includes a rating scale for the measures, list of objectives (with spaces to indicate rating and milestones when complete), list of competencies with ratings, summaries of key strengths and areas for improvement.

Recruitment, Selection and Orientation

An interview guide based on the critical incident interview used to identify competencies may support the hiring and selection process. The interview will explore past behaviors of candidates, based on the principle that the best predictor of future performance is past performance. Instruments for rating the candidates based on competencies indicated, as well as behaviors exhibited during the interview, also accompany the interview guide.

Other possible recruitment, selection, and orientation deliverables include:

- standards for ads
- template and procedures for job posting
- form communications to candidates
- procedures for offer letters
- pre-selection procedures and criteria
- interviewing procedures and guides
- procedures for resumes, solicited and unsolicited
- procedures for the use of recruiting agencies
- selection matrices
- interview logs
- orientation materials and resource list: 1st week, 1st month, 1st year
- hiring procedures
- tips on interviewing techniques, including legal do's and don'ts

Termination and Outplacement

Termination interview guides will cover the legal implications of the process, in addition to the leading practices for conducting the meeting. A checklist for termination includes the following considerations:

- reasons for termination
- documentation and performance appraisals
- content of communications to the employee
- severance benefits/arrangements (salary, health coverage, money stock purchase plans, 401K, etc.)
- outplacement options
- tips on interview techniques (sensitivity, unequivocal communication, listening, etc.)
- follow-up processes (letter, etc.)
- timing (not Friday, to allow for follow-up if needed; check for compliance with government regulation, for example, requiring xx days notice before closure of a plant or workplace, etc.)
- logistics (return of keys, removal of personal property, inventory badges, etc.)

Outplacement resources include lists of referrals to outside consultants or agencies the organization uses to assist in the outplacement process, skills and interest self-assessment instruments, resume writing and editing support, etc.

Rewards and Recognition Programs

Rewards and recognition deliverables include data sheets explaining how the programs work, guidelines for determining eligibility and payout, procedures, policies, certificate templates, etc.

New Information Technology Job-Person Matching

Job-person matching deliverables may include:

- Mapping skills profiles to skills requirements
- Learning plan
- Succession planning procedures

Information Technology Human Performance Support Systems Measurements

The general goal of customizing instruments for evaluating performance support effectiveness is to create materials that will effectively measure the performance support objectives, to determine the effectiveness of the performance support materials and approaches in meeting them, and to allow for revisions and recommendations for ongoing support.

Information Technology Learning Plan

This component contains a Learning Plan which recommends the most appropriate approaches and curriculum to re-skill the Information Technology groups' employees. It includes recommendations for reinforcement mechanisms, reward and recognition processes, and measurement methods.

Objectives

The objectives of the Information Technology Learning Plan are as follows:

- Propose a learning approach which meets the needs of the Information Technology groups' employees by customizing the learning objectives, learning paths, and learning components.
- Recommend an approach which enables learners to achieve the performance goals necessary for the Information Technology groups to achieve the business results anticipated from its new mission.
- Recommend a reinforcement mechanism, as well as rewards and recognition process, that makes sure of the transfer of newly acquired skills to on-the-job performance.
- Recommend valid methods to measure the effectiveness of the learning.
- Identify and remove any barriers caused by organizational, environmental, or logistical issues.

Learning Needs Analysis

The Learning Needs Analysis consisted of a series of interviews with a cross-section of the Information Technology groups' managers and employees. These respondents identified the job roles in their groups, described the gaps between the needed competencies and the current competencies for each new role, and then did the same thing for their own new roles. They also shared ideas for the Learning Plan particularly in the areas of delivery options, rewards and recognition, reinforcement and measurement. The respondents identified gaps in knowledge, skills, and abilities for each role and added further detail by supplying that information for each area of competency.

Learning Objectives per Role

All learning is aimed at improving performance whether it involves providing employees with knowledge they don't currently have, skills they have yet to master, or motivating them to make better use of their abilities. Learning Objectives or Learning Outcomes based on the results of the Learning Needs Analysis provide a clear direction for learning.

A Learning Objective describes what the learner will be able to do at the conclusion of the learning. The format for a Learning Objective includes three elements which provide a framework for the content: performance, condition, and criteria. These elements answer three questions:

- What should the learner be able to do?

- Under what conditions should the learner be able to do it?
- How well must it be done?

Learning Objectives/Role			
Role / (Incumbent)	Performance	Condition	Criteria
DBA			
System Manager			
System Operator			
Security Manager			
Network Manager			
Help Desk Analyst			
System Administrator			
Network Administrator			
System Architect			
Developer/ Analyst			
System Analyst/Engineer			
Technical Writer			
Data Center Manager			
Hardware Technician			
Database Designer			
System Developer			
Network Engineer			
Communications Engineer			
Web Administrator/master			

Learning Content

The Learning Needs Analysis gives us a starting point for creating a learning approach by identifying target groups of learners and the gaps in their knowledge, skills, and abilities. The Learning Objectives defined the ending point that is, where we want to be when the learning is achieved. The learning content or the curriculum structure forms the bridge between the two by defining the what it is that people need to know, or learn, or do in order to reach those objectives. The content should be role specific.

Learning Content			
Role / (Incumbent)	Mandatory	Strongly Recommended	Optional
DBA			
System Manager			
System Operator			
Security Manager			
Network Manager			
Help Desk Analyst			
System Administrator			
Network Administrator			
System Architect			
Developer/ Analyst			
System Analyst/Engineer			
Technical Writer			
Data Center Manager			

Learning Content			
Role / (Incumbent)	Mandatory	Strongly Recommended	Optional
Hardware Technician			
Database Designer			
System Developer			
Network Engineer			
Communications Engineer			
Web Administrator/master			

Learning Approach and Delivery Strategy per Target Group/Role

This Learning Approach reflects the optimal methods for transferring the needed skills to each group of learners, given their unique needs and the constraints or the opportunities provided by the environment. This focuses on the full range of the learning experience that is, the preparation, the support offered during learning, and the follow up when the learners return to the job. The follow up or post learning activities include learning reinforcement methods to support the transfer and maintenance of skills acquired to the learner's job environment. In addition, the media selection reflects the logistical requirements, to include number of location, number of learners, budget, turnaround time, available resources, etc.

Selecting the Media

The media have the following characteristics:

- Seem most likely to produce the expected learning outcomes
- Fit the learning needs of the target group of learners
- Mirror real work conditions and environment
- Are cost justified
- Fit the learners' culture
- Are practical to package, implement, and update
- Fit the organization's culture and history

Types of Learning Media Considered

The following captures the learning media selected:

Types of Media	Learning Media
Information: the learner is not held responsible for specific actions or measurable performance.	Conference Call E-mail Memo On-Line Message
Instructor-Led Learning	Case Method Demonstration Lecture/Presentation One-on-one Coaching Workshop and Labs
Paper Based Learning	Job Aids Desk Top Manual

Types of Media	Learning Media
Technology Based Learning	Video Based Computer Based Web Based Audio Based Interactive Distance Learning Multimedia Based (CD-ROM/CD)

Learning Paths per Role

The Learning Path combines the content, the media, and the sequence to create a flow of learning events for each targeted learning group. The following shows a sample learning path for the role of a database administrator.

Role: Database Administrator	Learning Path
Pre Learning Activities	Email messages "What to Expect?" Hands On Demo Preview in the Learning Lab Introductory Computer Based Learning
Learning Events as part of the Professional Path for ...	Instructor-Led Learning and Lab Exercise on: ~ ... Computer Based Learning
Post-Learning Events	Learner Verification/Certification Serving as a coach for a new team member

Learning Rewards and Recognition Process

Everyone wants to be recognized for accomplishments and rewarded for increased productivity and contribution to the success of the organization for which they work. A proven tool for motivating employee performance is recognition for a job well done, and the effect builds upon itself. A job well done which is recognized provides an incentive for yet more productivity and improved performance.

- Recognition during learning is best when it is informal, fun, spontaneous, varied, culturally sensitive and provides equal opportunities for everyone. We suggest the following:...
- Recognition given at the end of learning is best when it is formal, public (in Western cultures), and matched to achievement. We suggest the following: ...
- Recognition/rewards given on the job, following learning is best when personal and tailored to the individual, timely, cumulative and matched to the achievement. In this case, successful completion of the learning (and certification from the Professional Paths) can be linked to career development. We suggest the following:

Learning Measurement Methods

The general goal of developing learning measurement methods is to create a means of gauging the success of the learning and shape recommendations for ongoing learning. We are using learner verification and successful completion of the

Professional Development Paths (certification) as the means to measure the effectiveness of the learning.

Learning Events Schedule

This following table outlines the schedule for the Information Technology individual learning events.

Date	Name	Job Role/Function	Event Title	Prerequisites	Location

Roles and Responsibilities

The following people are involved in the successful completion of the Information Technology groups' Learning Plan

Names	Responsibilities
	Signs off on the Information Technology groups' learning plan
	Communicates about the Information Technology groups' learning plan
	Provides funding and other required resources
	Confirms logistics are in place
	Tracks administration of learning events
	Progress reports
	Provides rewards and recognition

Learning Events Tracking Mechanism

We will use the following system to track the learning events and their participants:

Contact Information

Learning Coordinator(s)/Contacts

The following documents information on the learning coordinator:

Information Technology Group	Learning Coordinator	Address	Phone

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Appendix A - Common Information Technology Job Roles

The following lists common information technology roles and the related Oracle products:

Common Information Technology Roles	Related Oracle Products
Administrator Database Administrator Network Administrator On-Line Analytical Processing (OLAP) Administrator Platform Specialist (H, Solaris, Microsoft) Internet DBA Oracle Applications Administrator Decision Support Systems (DSS) Administrator Data Access Administrator	Database Express Discoverer Oracle Applications Server Oracle Developer Server Enterprise Manager
Design Architect Designer Data Administrator System Analyst Application Architect Program Analyst Application Architect Program Analyst	Oracle Designer Enterprise Developer Suite Oracle Applications
Developer Enterprise Database Application Developer - Oracle Developer Application Developer - 3 GL Database Developer - Oracle Applications Developer Java and Internet Applications Developer Web Publisher Developer	Oracle Developer Oracle Application Server Enterprise Developer Suite Jdeveloper Suite Site Developer Suite Oracle Applications Oracle WebDB
Help Desk Manager Supervisor Specialist Intern	

Appendix B - Traits for Information Technology Competencies

The following lists traits for information technology competencies:

- Define the architecture and processes of the Oracle Relational Database Server
- Use interactive SQL to issue DML and DDL commands
- Create and manage the database, load data, open and close the database, grant and revoke privileges, backup and restore the database, restructure the database and monitor and evaluate performance
- Measure database performance using diagnostic tools and apply tuning techniques to improve performance
- Define and implement a physical database design
- Define the architecture and processes of the Oracle server
- Create and manage the database from startup through shutdown, granting privileges, performing backup and recovery and monitoring performance
- Manage data integrity constraints, manage indices and clusters and monitor lock contention
- Configure the system for heterogeneous network environments when additional software options are purchased
- Configure, tune and troubleshoot Oracle SQL*Net in a client/server environment
- Use Oracle Express and Oracle Express Administrator to create and manage multidimensional databases that satisfy simulated business requirements
- Use Oracle Express Administrator to install and configure multidimensional databases for optimal use with client applications
- Use Oracle Express Objects to develop object-oriented OLAP applications combining event-driven programming with customized objects, dialogs, menus, toolbars, and interactive tables and graphs
- Use fundamental UNIX or NT commands
- Install and maintain Oracle on UNIX or NT
- Configure and manage the UNIX or NT system
- Administer the Oracle server on your chosen UNIX or NT platform
- Install Oracle Web Application Server and configure the Web Listener, Web Request Broker, DCDs and standard cartridges
- Set up an Internet set using Oracle Web Application Server
- Configure Oracle SQL*Net server
- Prepare the Oracle server to deploy web-enabled applications
- Administer and use the Web Request Broker and the standard cartridges
- Put in place Internet security

- Use SQL and PL/SQL in the development of Oracle Web Applications Server applications
- Install Oracle Web Applications Server and configure the Web Listener, Web Request Broker, DCDs and standard cartridges
- Build Static HTML pages and dynamic web-enabled applications using the Software Developer's Toolkit
- Build application cartridges to provide custom web services
- Use SQL and PL/SQL in the development of applications
- Build form applications using the Form Builder
- Build efficient multiple form applications
- Build reports using the Report Builder
- Build reports remotely using the Report Server
- Manage applications using the Project Builder
- Build complete Developer applications incorporating forms, reports and charts
- Incorporate OCX Controls in a Developer application
- Build effective and consistent Graphical User Interface Oracle Applications Master Extension for Application Developer, Developer Masters
- Understand Oracle Applications product functions
- Build or tailor applications using Oracle Application Object Library
- Understand the technical elements of Oracle Financial or Oracle Manufacturing Applications
- Communicate effectively between the database administration staff and the development staff
- Identify the basic Oracle architecture, database design concepts, data integrity constraint implementation and security issues as they relate to the implementation of applications
- Develop and manage Oracle stored procedures and database triggers
- Manage exception handling using programming constructs
- Define efficient use of SQL in various application methods
- Use diagnostic and performance tools to tune applications
- Define system scope and functionality and manage user expectations
- Create detailed information and process models of user and business requirements
- Prototype and reverse-engineer to enhance business system models
- Identify requirements for acceptance testing, conversion and learning
- Understand analysis techniques to read and interpret business models
- Define logical and physical database schemas
- Develop program/modules structures
- Understand database sizing, index requirements and performance and optimization issues

- Create executable code required to build and implement systems
- Understand the architecture and processes of the Oracle Server
- Manage the database through startup and shutdown, backup and recovery, granting privileges and maintaining audit trails
- Understand the security and administration of Oracle Applications
- Identify the basic architecture underlying Oracle Applications, including critical tables, indices and views
- Perform archiving and purging functions provided by Oracle Applications
- Configure Oracle data and log files for optimal performance
- Understand relational database design concepts
- Use SQL and PL/SQL, Developer and Application Object Library in the development and maintenance of Oracle Applications
- Support the product division(s) in the daily technical administration of Oracle Applications
- Perform critical setup functions on Oracle Applications
- Understand the security and software administration of Oracle Applications
- Use Oracle Applications product features to manage business functions
- Understand the functionality of each of Oracle's Manufacturing Applications
- Understand the functionality of each of Oracle's Financial Applications
- Set up and perform activities necessary to support the human resources management function in the organization
- Navigate through Oracle Applications using the graphical user interface (GUI)
- Understand how to enter data, search for data, customize data presentations and use online help in Oracle Applications
- Understand how Flexbuilder is used to customize Oracle Applications
- Set up, maintain and test Flexbuilder rules
- Define conditions to meet specific needs
- Understand multiple organizational (Multi-Organizational) features in Oracle Applications
- Match enterprise needs to Multi-Organizational features
- Develop the organizational structure in Oracle Applications
- Program in COBOL, C, Fortran, C++, Visual Basic, HTML, PASCAL, Java
- Program for Sybase, Informix, Ingres, DB/2, Rdb, Access, FoxPro, Adabas, Approach, Dbase, Paradox, VSAM, IDMS, RedBrick